

On estimation of the noise variance in high-dimensional probabilistic principal component analysis

(Supplementary report)

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1 Figure for Section 2.1

Figure 1 presents the histograms from 1000 replications of

$$\frac{(p-m)}{\sigma^2\sqrt{2c_n}}(\hat{\sigma}^2 - \sigma^2) + b(\sigma^2)$$

of the three models in Section 2.1, with sample size $n = 100$ and dimensions $p = c \times n$, compared to the density of the standard Gaussian distribution. The sampling distribution is almost normal.

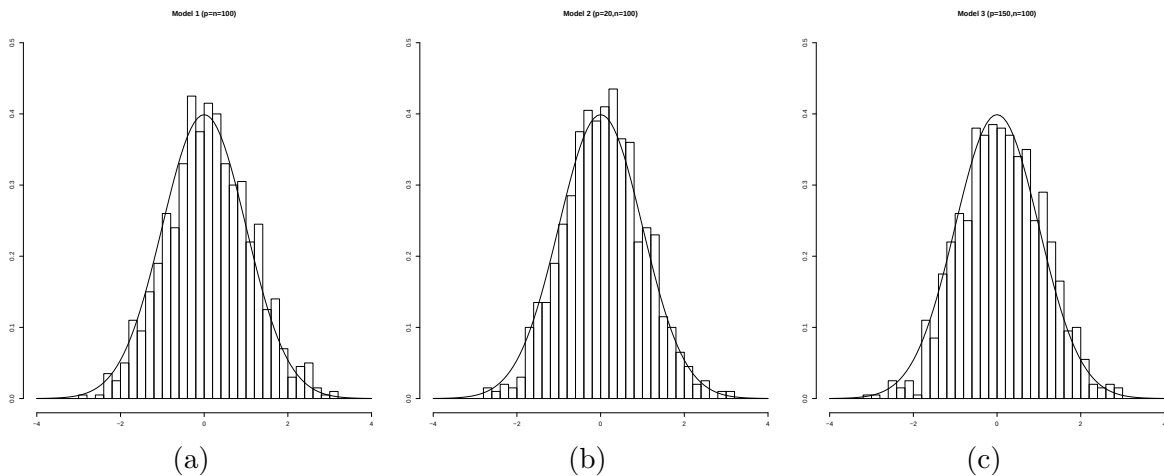


Figure 1. Histogram of $\frac{(p-m)}{\sigma^2\sqrt{2c}}(\hat{\sigma}^2 - \sigma^2) + b(\sigma^2)$ compared with the density of a standard Gaussian distribution.

2 More Monte-Carlo experiments for Section 2.3

Tables 1 and 2 report the empirical means of the estimator of the number of PCs over 1000 replications, for $m = 3$ and 5 respectively, with standard errors in parentheses. When

a standard error is actually zero, no standard error is thus indicated. For all cases, the predetermined maximum number m_0 of PCs is set to 8.

3 Application to the SURE criterion

Ulfarsson and Solo (2008) proposes to use the SURE criteria to choose the number of PCs. This criterion uses the noise variance estimator $\hat{\sigma}_{US}^2$ defined in Section 2. It aims at minimizing the Euclidean distance between the underlying estimator of the population mean $\boldsymbol{\mu}$ and its true value. The proposed SURE criterion for m number of PCs (to be minimized) is

$$\begin{aligned} R_m = & (p-m)\hat{\sigma}_{US}^2 + \hat{\sigma}_{US}^4 \sum_{j=1}^m \frac{1}{\lambda_j} + 2\hat{\sigma}_{US}^2(1-1/n)m \\ & - 2\hat{\sigma}_{US}^4(1-1/n) \sum_{j=1}^m \frac{1}{\lambda_j} + \frac{4(1-1/n)\hat{\sigma}_{US}^4}{n} \sum_{j=1}^m \frac{1}{\lambda_j} + C_m, \end{aligned} \quad (1)$$

where

$$\begin{aligned} C_m = & \frac{4(1-1/n)\hat{\sigma}_{US}^2}{n} \sum_{j=1}^m \sum_{i=m+1}^p \frac{\lambda_j - \hat{\sigma}_{US}^2}{\lambda_j - \lambda_i} + \frac{2(1-1/n)\hat{\sigma}_{US}^2}{n} m(m-1) \\ & - \frac{2(1-1/n)\hat{\sigma}_{US}^2}{n} (p-1) \sum_{j=1}^m \left(1 - \frac{\hat{\sigma}_{US}^2}{\lambda_j}\right). \end{aligned}$$

Recall that $\hat{\sigma}_{US}^2$ is also related to m . From Section 2, we have known that $\hat{\sigma}_{US}^2$ is not as good as our bias-corrected estimator. To examine further this difference, we replace $\hat{\sigma}_{US}^2$ with $\hat{\sigma}_*^2$ in (11), referred then as SURE*, to see whether the performance of SURE can be improved.

Then simulation experiments are conducted to check the performance of SURE*. The setup follows the paper Ulfarsson and Solo (2008) and the data are simulated according to (1) with the parameters $p = 64, p/n = [2/3, 1/2, 2/5], m = [5, 10, 15, 20]$ and $\sigma^2 = 1$. The loading matrix is set to $\mathbf{\Lambda} = \mathbf{F}\mathbf{D}^{1/2}$, where $\mathbf{F}$ is constructed by generating a $p \times m$ matrix of Gaussian random variables and then orthogonalizing the resulting matrix, and $\mathbf{D} = \text{diag}((m+1)^2, m^2, \dots, 3^2, \lambda_m), \lambda_m = 1.5$. All simulations were repeated 1500 times. Table 3 presents the percentage of correct selection of number of PCs for SURE and SURE*, and the results of SURE are from Table II of Ulfarsson and Solo (2008). SURE* largely outperforms SURE in all of the tested cases, most of times by a wide margin. All the percentages of correct selection of SURE* are larger than 90% and in 4 out of 12 cases, the

Table 1: Comparison between PC_p^* s and PC_p s in terms of the mean estimation numbers of PCs for $m = 3, \theta = 3$.

N	T	PC_{p1}^*	PC_{p2}^*	PC_{p3}^*	PC_{p1}	PC_{p2}	PC_{p3}
100	40	2.98(0.15)	2.95(0.22)	3.00(0.06)	3.00	3.00	3.90
100	60	3.00(0.03)	3.00(0.04)	3.00	3.01(0.08)	3.00	4.37(0.64)
200	60	3.00	3.00	3.00	3.00	3.00	4.18(0.63)
500	60	3.00	3.00	3.00	3.00	3.00	3.00
1000	60	3.00	3.00	3.00	3.00	3.00	3.00
2000	60	3.00	3.00	3.00	3.00	3.00	3.00
100	100	3.00	3.00	3.00	3.00	3.00	5.62(0.72)
200	100	3.00	3.00	3.00	3.00	3.00	3.00
500	100	3.00	3.00	3.00	3.00	3.00	3.00
1000	100	3.00	3.00	3.00	3.00	3.00	3.00
2000	60	3.00	3.00	3.00	3.00	3.00	3.00
40	100	2.99(0.10)	2.98(0.14)	3.00	3.07(0.26)	3.01(0.07)	5.04(0.72)
60	100	3.00	3.00(0.03)	3.00	3.00	3.00	4.65(0.69)
60	200	3.00	3.00	3.00	3.00	3.00	3.00
60	500	3.00	3.00	3.00	3.00	3.00	3.00
60	1000	3.00	3.00	3.00	3.00	3.00	3.00
60	2000	3.00	3.00	3.00	3.00	3.00	3.00
4000	60	3.00	3.00	3.00	3.00	3.00	3.00
4000	100	3.00	3.00	3.00	3.00	3.00	3.00
8000	60	3.00	3.00	3.00	3.00	3.00	3.00
8000	100	3.00	3.00	3.00	3.00	3.00	3.00
60	4000	3.00	3.00	3.00	3.00	3.00	3.00
100	4000	3.00	3.00	3.00	3.00	3.00	3.00
60	8000	3.00	3.00	3.00	3.00	3.00	3.00
100	8000	3.00	3.00	3.00	3.00	3.00	3.00
10	50	8.00	8.00	8.00	8.00	8.00	8.00
10	100	8.00	8.00	8.00	8.00	8.00	8.00
20	100	2.89(0.32)	2.85(0.37)	2.95(0.27)	6.55(0.74)	5.96(0.77)	7.62(0.55)
100	10	2.57(1.35)	2.43(1.19)	2.77(1.54)	8.00	8.00	8.00
100	20	2.46(0.63)	2.37(0.65)	2.65(0.52)	6.15(0.69)	5.46(0.68)	7.49(0.59)

Table 2: Comparison between PC_p^* s and PC_p s in terms of the mean estimation numbers of PCs for $m = 5, \theta = 5$.

N	T	PC_{p1}^*	PC_{p2}^*	PC_{p3}^*	PC_{p1}	PC_{p2}	PC_{p3}
100	40	3.83(0.77)	3.49(0.77)	4.51(0.58)	5.00(0.07)	4.98(0.15)	5.36(0.51)
100	60	4.66(0.50)	4.36(0.61)	4.98(0.13)	5.00(0.03)	5.00(0.06)	5.27(0.45)
200	60	4.95(0.22)	4.90(0.30)	4.99(0.08)	5.00	5.00	5.00
500	60	5.00(0.04)	5.00(0.07)	5.00(0.03)	5.00	5.00	5.00
1000	60	5.00(0.04)	5.00(0.04)	5.00	5.00	5.00	5.00
2000	60	5.00(0.03)	5.00(0.03)	5.00(0.03)	5.00	5.00	5.00
100	100	4.(0.12)	4.90(0.30)	5.00	5.00	5.00	6.18(0.63)
200	100	5.00	5.00	5.00	5.00	5.00	5.00
500	100	5.00	5.00	5.00	5.00	5.00	5.00
1000	100	5.00	5.00	5.00	5.00	5.00	5.00
2000	60	5.00	5.00	5.00	5.00	5.00	5.00
40	100	4.25(0.68)	3.92(0.75)	4.77(0.44)	4.98(0.04)	5.66(0.14)	5.66(0.57)
60	100	4.76(0.44)	4.47(0.60)	4.76(0.10)	5.00(0.03)	4.99(0.08)	5.46(0.56)
60	200	4.97(0.17)	4.94(0.24)	5.00	5.00	5.00	5.00
60	500	5.00(0.05)	5.00(0.06)	5.00(0.04)	5.00	5.00	5.00
60	1000	5.00(0.03)	5.00(0.03)	5.00	5.00	5.00	5.00
60	2000	5.00	5.00	5.00	5.00	5.00	5.00
4000	60	5.00	5.00	5.00	5.00	5.00	5.00
4000	100	5.00	5.00	5.00	5.00	5.00	5.00
8000	60	5.00	5.00	5.00	5.00	5.00	5.00
8000	100	5.00	5.00	5.00	5.00	5.00	5.00
60	4000	5.00	5.00	5.00	5.00	5.00	5.000
100	4000	5.00	5.00	5.00	5.00	5.00	5.00
60	8000	5.00	5.00	5.00	5.00	5.00	5.00
100	8000	5.00	5.00	5.00	5.00	5.00	5.00
10	50	8.00	8.00	8.00	8.00	8.00	8.00
10	100	8.00	8.00	8.00	8.00	8.00	8.00
20	100	3.64(0.91)	3.38(0.94)	4.08(0.79)	6.65(0.64)	6.12(0.64)	7.63(0.51)
100	10	3.10(2.01)	2.83(1.86)	3.53(2.27)	8.00	8.00	8.00
100	20	2.18(0.92)	1.93(0.92)	2.65(0.0.90)	6.56(0.62)	5.97(0.62)	7.66(0.50)

Table 3: Comparison between SURE and SURE* in terms of percentage of correct selection of PCs.

m		$p/n = 2/3$	$p/n = 1/2$	$p/n = 2/5$
5	SURE*	1.000	1.000	1.000
	SURE	0.408	0.621	0.807
10	SURE*	0.992	1.000	0.998
	SURE	0.512	0.739	0.858
15	SURE*	0.920	0.978	0.989
	SURE	0.598	0.783	0.911
20	SURE*	0.909	0.966	0.990
	SURE	0.617	0.810	0.899

detection rate is 100%. Therefore, by implementing our bias-corrected estimator of the noise variance instead of the one provided by its authors, the SURE criterion has a much better performance.

4 Proofs

Before giving the proofs, we first recall some important results from the random matrix theory which laid the foundation for the proofs of the main results of the paper.

4.1 Useful results from random matrix theory

Random matrix theory has become a powerful tool to address new inference problems in high-dimensional scheme. For general background and references, we refer to review papers Johnstone (2007) and Johnstone and Titterton (2009).

Let H be a probability measure on $\mathbb{R}^+$ and $c > 0$ a constant. We define the map

$$g(s) = g_{c,H}(s) = \frac{1}{s} + c \int \frac{t}{1+ts} dH(t) \quad (2)$$

in the set $\mathbb{C}^+ = \{z \in \mathbb{C} : \Im z > 0\}$. The map g is a one-to-one mapping from $\mathbb{C}^+$ onto itself (see Bai and Silverstein, 2010, Chapter 6), and the inverse map $m = g^{-1}$ satisfies all the requirements of the Stieltjes transform of a probability measure on $[0, \infty)$. We call this measure $\underline{F}_{c,H}$. Next, a companion measure $F_{c,H}$ is introduced by the equation $cF_{c,H} =$

$(c - 1) \delta_0 + F_{c,H}$ (note that in this equation, measures can be signed). The measure $F_{c,H}$ is referred as the generalized Marčenko-Pastur distribution with index (c, H) .

Let $F_n = \frac{1}{p} \sum_{i=1}^p \delta_{\lambda_{n,i}}$ be the empirical spectral distribution (ESD) of the sample covariance matrix $\mathbf{S}_n$ defined in (2) in main paper with the $\{\lambda_{n,i}\}$ denoting its eigenvalues. Then, it is well-known that under suitable moment conditions, F_n converges to the Marčenko-Pastur distribution of index (c, δ_{σ^2}) , simply denoted as F_{c,σ^2} , with the following density function

$$p_{c,\sigma^2}(x) = \begin{cases} \frac{1}{2\pi x c \sigma^2} \sqrt{\{b(c) - x\}\{x - a(c)\}}, & a(c) \leq x \leq b(c), \\ 0, & \text{otherwise.} \end{cases}$$

The distribution has an additional mass $(1 - 1/c)$ at the origin if $c > 1$.

The ESD H_n of $\mathbf{\Sigma}$ is

$$H_n = \frac{p-m}{p} \delta_{\sigma^2} + \frac{1}{p} \sum_{i=1}^m \delta_{\alpha_i + \sigma^2},$$

and $H_n \rightarrow \delta_{\sigma^2}$. Define the normalized empirical process

$$G_n(f) = p \int_{\mathbb{R}} f(x) [F_n - F_{c_n, H_n}](dx), \quad f \in \mathcal{A},$$

where $\mathcal{A}$ is the set of analytic functions $f : \mathcal{U} \rightarrow \mathbb{C}$, with $\mathcal{U}$ an open set of $\mathbb{C}$ such that $[1_{(0,1)}(c)a(c), b(c)] \subset \mathcal{U}$. We will need the following CLT which is a combination of Theorem 1.1 of Bai and Silverstein (2004) and a recent addition proposed in Zheng et al. (2015).

Proposition 1. *We assume the same conditions as in Theorem 1. Then, for any functions $f_1, \dots, f_k \in \mathcal{A}$, the random vector $(G_n(f_1), \dots, G_n(f_k))$ converges to a k -dimensional Gaussian vector with mean vector*

$$m(f_j) = \frac{f_j(a(c)) + f_j(b(c))}{4} - \frac{1}{2\pi} \int_{a(c)}^{b(c)} \frac{f_j(x)}{\sqrt{4c\sigma^4 - (x - \sigma^2 - c\sigma^2)^2}} dx, \quad j = 1, \dots, k,$$

and covariance function

$$v(f_j, f_l) = -\frac{1}{2\pi^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} \frac{f_j(z_1) f_l(z_2)}{(\underline{m}(z_1) - \underline{m}(z_2))^2} d\underline{m}(z_1) d\underline{m}(z_2), \quad j, l = 1, \dots, k, \quad (3)$$

where $\underline{m}(z)$ is the Stieltjes transform of $F_{c,\sigma^2} = (1 - c)\delta_0 + cF_{c,\sigma^2}$. The contours $\mathcal{C}_1$ and $\mathcal{C}_2$ are non overlapping and both contain the support of F_{c,σ^2} .

An important and subtle point here is that the centering term in $G_n(f)$ in the above CLT is defined with respect to the Marčenko-Pastur distribution F_{c_n, H_n} with “current” index

(c_n, H_n) instead of the limiting distribution F_{c, σ^2} with index (c, σ^2) . In contrast, the limiting mean function $m(f_j)$ and covariance function $v(f_j, f_l)$ depend on the limiting distribution F_{c, σ^2} only.

4.2 Proof of Proposition 1 in main paper

By Lemma 2.2 of Wang and Yao (2013),

$$\sum_{i=1}^p \lambda_i^2 - p \int x^2 dF_{c_n, H_n}(x) \xrightarrow{\mathcal{D}} \mathcal{N}(m(x^2), v),$$

with $m(x^2) = c\sigma^4(\gamma - 1)$ and some computable $v > 0$. Furthermore, by Lemma 1 of Bai et al. (2010),

$$\int x^2 dF_{c_n, H_n}(x) = \beta_2 + \frac{p}{n} \beta_1^2,$$

where

$$\beta_1 = \sigma^2 + \frac{1}{p} \sum_{j=1}^m \alpha_j, \text{ and } \beta_2 = \sigma^4 + \frac{1}{p} \sum_{j=1}^m \alpha_j^2 + \frac{2}{p} \sigma^2 \sum_{j=1}^m \alpha_j.$$

The conclusion follows.

4.3 Proof of Theorem 6 in main paper

We have

$$\begin{aligned} L^* &= \sum_{i=m+1}^p \log \frac{\lambda_{n,i}}{\hat{\sigma}^2} \\ &= \sum_{i=m+1}^p \log \frac{\lambda_{n,i}}{\sigma^2} - \sum_{i=m+1}^p \log \frac{\hat{\sigma}^2}{\sigma^2} \\ &= \sum_{i=m+1}^p \log \frac{\lambda_{n,i}}{\sigma^2} - (p-m) \log \left(\frac{1}{p-m} \sum_{i=m+1}^p \frac{\lambda_{n,i}}{\sigma^2} \right) \\ &= L_1 - (p-m) \log \left(\frac{L_2}{p-m} \right), \end{aligned}$$

where we have defined a two-dimensional vector $(L_1, L_2) = (\sum_{i=m+1}^p \log \frac{\lambda_{n,i}}{\sigma^2}, \sum_{i=m+1}^p \frac{\lambda_{n,i}}{\sigma^2})$.

CLT when $\sigma^2 = 1$. To start with, we consider the case $\sigma^2 = 1$. We have

$$\begin{aligned} L_1 &= p \int \log(x) dF_n(x) - \sum_{i=1}^m \log \lambda_{n,i} \\ &= p \int \log(x) d(F_n - F_{c_n, H_n})(x) + p \int \log(x) dF_{c_n, H_n}(x) - \sum_{i=1}^m \log \lambda_{n,i}. \end{aligned}$$

Similarly, we have

$$L_2 = p \int x d(F_n - F_{c_n, H_n})(x) + p \int x dF_{c_n, H_n}(x) - \sum_{i=1}^m \lambda_{n,i}.$$

By Proposition 1, we find that

$$p \begin{pmatrix} \int \log(x) d(F_n - F_{c_n, H_n})(x) \\ \int x d(F_n - F_{c_n, H_n})(x) \end{pmatrix} \xrightarrow{\mathcal{D}} \mathcal{N} \left(\begin{pmatrix} m_1(c) \\ m_2(c) \end{pmatrix}, \begin{pmatrix} v_1(c) & v_{1,2}(c) \\ v_{1,2}(c) & v_2(c) \end{pmatrix} \right) \quad (4)$$

with $m_2(c) = 0$ and $v_2(c) = 2c$ and

$$m_1(c) = \frac{\log(1-c)}{2}, \quad (5)$$

$$v_1(c) = -2\log(1-c), \quad (6)$$

$$v_{1,2}(c) = 2c. \quad (7)$$

Formulae of m_2 and v_2 have been established in the proof of Theorem 1 and the others are derived in next subsection.

In Theorem 1, with $\sigma^2 = 1$, we found that

$$\int x dF_{c_n, H_n}(x) = 1 + \frac{1}{p} \sum_{i=1}^m \alpha_i,$$

and

$$\sum_{i=1}^m \lambda_{n,i} \xrightarrow{\text{a.s.}} \sum_{i=1}^m \left(\alpha_i + \frac{c}{\alpha_i} \right) + m(1+c).$$

For the last term of L_1 , by (6) in main paper, we have

$$\log \lambda_{n,i} \longrightarrow \log(\phi(\alpha_i + 1)) = \log((\alpha_i + 1)(1 + c\alpha_i^{-1})) \text{ a.s.}$$

Furthermore, by Wang et al. (2014), we have

$$\int \log(x) dF_{c_n, H_n}(x) = \frac{1}{p} \sum_{i=1}^m \log(\alpha_i + 1) + h(c_n) + o\left(\frac{1}{p}\right),$$

where

$$h(c_n) = \int \log(x) dF_{c_n, \delta_1}(x) = \frac{c_n - 1}{c_n} \log(1 - c_n) - 1. \quad (8)$$

can be calculated using the density of the Marčenko-Pastur law (see 4.1). Summarising, we have obtained that

$$L_1 - m_1(c) - ph(c_n) + \eta(c, \alpha) \xrightarrow{\mathcal{D}} \mathcal{N}(0, v_1(c)),$$

where $h(c_n) = \frac{c_n - 1}{c_n} \log(1 - c_n) - 1$ and $\eta(c, \alpha) = \sum_{i=1}^m \log(1 + c\sigma^2 \alpha_i^{-1})$. Similarly, we have

$$L_2 - (p - m) + \rho(c, \alpha) \xrightarrow{\mathcal{D}} \mathcal{N}(0, v_2(c)),$$

where $\rho(c, \alpha) = c(m + \sum_{i=1}^m \alpha_i^{-1})$.

Using (4) and Slutsky's lemma,

$$\begin{pmatrix} L_1 \\ L_2 \end{pmatrix} \xrightarrow{\mathcal{D}} \mathcal{N}\left(\begin{pmatrix} m_1(c) + ph(c_n) - \eta(c, \alpha) \\ p - m - \rho(c, \alpha) \end{pmatrix}, \begin{pmatrix} v_1(c) & v_{1,2}(c) \\ v_{1,2}(c) & v_2(c) \end{pmatrix}\right),$$

with $h(c_n) = \frac{c_n - 1}{c_n} \log(1 - c_n) - 1$, $\eta(c, \alpha) = \sum_{i=1}^m \log(1 + c\sigma^2 \alpha_i^{-1})$ and $\rho(c, \alpha) = c(m + \sum_{i=1}^m \alpha_i^{-1})$.

CLT with general σ^2 . When $\sigma^2 = 1$,

$$\text{spec}(\mathbf{\Sigma}) = (\alpha_1 + 1, \dots, \alpha_m + 1, 1, \dots, 1),$$

whereas in the general case

$$\begin{aligned} \text{spec}(\mathbf{\Sigma}) &= (\alpha_1 + \sigma^2, \dots, \alpha_m + \sigma^2, \sigma^2, \dots, \sigma^2) \\ &= \sigma^2 \left(\frac{\alpha_1}{\sigma^2} + 1, \dots, \frac{\alpha_m}{\sigma^2} + 1, \dots, 1 \right). \end{aligned}$$

Thus, if we consider λ_i/σ^2 , we will find the same CLT by replacing the $(\alpha_i)_{1 \leq i \leq m}$ by α_i/σ^2 . Furthermore, we divide L_2 by $p - m$ to find

$$\begin{pmatrix} L_1 \\ \frac{L_2}{p-m} \end{pmatrix} \xrightarrow{\mathcal{D}} \mathcal{N}\left(\begin{pmatrix} m_1(c) + ph(c_n) - \eta(c, \alpha/\sigma^2) \\ 1 - \frac{\rho(c, \alpha/\sigma^2)}{p-m} \end{pmatrix}, \begin{pmatrix} \frac{2c}{(p-m)^2} & \frac{2c}{p-m} \\ \frac{2c}{p-m} & -2\log(1 - c) \end{pmatrix}\right) \quad (9)$$

with $\eta(c, \alpha/\sigma^2) = \sum_{i=1}^m \log(1 + c\sigma^2\alpha_i^{-1})$, $\rho(c, \alpha/\sigma^2) = c(m + \sigma^2 \sum_{i=1}^m \alpha_i^{-1})$ and $h(c_n) = \frac{c_n-1}{c_n} \log(1 - c_n) - 1$.

Asymptotic distribution of L^* . We have $L^* = g(L_1, L_2/(p-m))$, with $g(x, y) = x - (p-m) \log(y)$. We will apply the multivariate delta-method on (9) with the function g . We have $\nabla g(x, y) = \left(1, -\frac{p-m}{y}\right)$ and

$$L^* \xrightarrow{\mathcal{D}} \mathcal{N}(\beta_1 - (p-m) \log(\beta_2), \nabla g(\beta_1, \beta_2) \text{cov}(L_1, L_2/(p-m)) \nabla g(\beta_1, \beta_2)'),$$

with $\beta_1 = m_1(c) + ph(c_n) - \eta(c, \alpha/\sigma^2)$ and $\beta_2 = 1 - \frac{\rho(c, \alpha/\sigma^2)}{p-m}$. After some standard calculation, we finally find

$$L^* \xrightarrow{\mathcal{D}} \mathcal{N}\left(m_1(c) + ph(c_n) - \eta\left(c, \frac{\alpha}{\sigma^2}\right) - (p-m) \log(\beta_2), -2 \log(1-c) + \frac{2c}{\beta_2} \left(\frac{1}{\beta_2} - 2\right)\right).$$

4.4 Complementary proofs

Proof of (4) in main paper

The general theory of the m.l.e. for the PPCA model (1) in the classical setting has been developed in Anderson and Amemiya (1988) with in particular the following result.

Proposition 2. *Let $\Theta = (\theta_{ij})_{1 \leq i, j \leq p} = \Psi - \Lambda(\Lambda' \Psi^{-1} \Lambda)^{-1} \Lambda'$. If $(\theta_{ij}^2)_{1 \leq i, j \leq p}$ is nonsingular, if Λ and Ψ are identified by the condition that $\Lambda' \Psi \Lambda$ is diagonal and the diagonal elements are different and ordered, if $\mathbf{S}_n \rightarrow \Lambda \Lambda' + \Psi$ in probability and if $\sqrt{n}(\mathbf{S}_n - \Sigma)$ has a limiting distribution, then $\sqrt{n}(\hat{\Lambda} - \Lambda)$ and $\sqrt{n}(\hat{\Psi} - \Psi)$ have a limiting distribution. The covariance of $\sqrt{n}(\hat{\Psi}_{ii} - \Psi_{ii})$ and $\sqrt{n}(\hat{\Psi}_{jj} - \Psi_{jj})$ in the limiting distribution is $2\Psi_{ii}^2 \Psi_{jj}^2 \xi^{ij}$ ($1 \leq i, j \leq p$), where $(\xi^{ij}) = (\theta_{ij}^2)^{-1}$.*

To prove the CLT (4) in main paper, by Proposition 2, we know that the inverse of the Fisher information matrix is $\mathcal{I}^{-1}(\psi_{11}, \dots, \psi_{pp}) = (2\psi_{ii}^2 \psi_{jj}^2 \xi^{ij})_{ij}$. We have to change the parametrization: in our case, we have $\psi_{11} = \dots = \psi_{pp}$. Let $g: \mathbb{R} \rightarrow \mathbb{R}^p$, $a \mapsto (a, \dots, a)$. The information matrix in this new parametrization becomes

$$\mathcal{I}(\sigma^2) = J' \mathcal{I}(g(\sigma^2)) J,$$

where J is the Jacobian matrix of g . As

$$\mathcal{I}(g(\sigma^2)) = \frac{1}{2\sigma^8} (\theta_{ij}^2)_{ij},$$

we have

$$\mathcal{I}(\sigma^2) = \frac{1}{2\sigma^8} \sum_{i,j=1}^p \theta_{ij}^2,$$

and

$$\begin{aligned} \Theta = (\theta_{ij})_{ij} &= \mathbf{\Psi} - \mathbf{\Lambda}(\mathbf{\Lambda}'\mathbf{\Psi}^{-1}\mathbf{\Lambda})^{-1}\mathbf{\Lambda}' \\ &= \sigma^2(\mathbf{I}_p - \mathbf{\Lambda}(\mathbf{\Lambda}'\mathbf{\Lambda})^{-1}\mathbf{\Lambda}'). \end{aligned}$$

By hypothesis, we have $\mathbf{\Lambda}'\mathbf{\Lambda} = \text{diag}(d_1^2, \dots, d_m^2)$. Consider the Singular Value Decomposition of $\mathbf{\Lambda}$, $\mathbf{\Lambda} = \mathbf{U}\mathbf{D}\mathbf{V}$, where $\mathbf{U}$ is a $p \times p$ matrix such that $\mathbf{U}\mathbf{U}' = \mathbf{I}_p$, $\mathbf{V}$ is a $m \times m$ matrix such that $\mathbf{V}'\mathbf{V} = \mathbf{I}_m$, and $\mathbf{D}$ is a $p \times m$ diagonal matrix with $d_1, \dots, d_m$ as diagonal elements. As $\mathbf{\Lambda}'\mathbf{\Lambda}$ is diagonal, $\mathbf{V} = \mathbf{I}_m$, so $\mathbf{\Lambda} = \mathbf{U}\mathbf{D}$. By elementary calculus, one can find that

$$\mathbf{\Lambda}(\mathbf{\Lambda}'\mathbf{\Lambda})^{-1}\mathbf{\Lambda}' = \text{diag}(\underbrace{1, \dots, 1}_m, \underbrace{0, \dots, 0}_{p-m}),$$

so

$$\Theta = \sigma^2 \text{diag}(\underbrace{0, \dots, 0}_m, \underbrace{1, \dots, 1}_{p-m}).$$

Finally,

$$\mathcal{I}(\sigma^2) = \frac{1}{2\sigma^8} (p-m)\sigma^4 = \frac{p-m}{2\sigma^4},$$

and the asymptotic variance of $\hat{\sigma}^2$ is

$$s^2 = \mathcal{I}^{-1}(\sigma^2) = \frac{2\sigma^4}{p-m}.$$

Proof of (17) in main paper

By Proposition 1, for $g(x) = x$, by using the variable change $x = \sigma^2(1 + c - 2\sqrt{c}\cos\theta)$, $0 \leq \theta \leq \pi$, we have

$$\begin{aligned} m(g) &= \frac{g(a(c)) + g(b(c))}{4} - \frac{1}{2\pi} \int_{a(c)}^{b(c)} \frac{x}{\sqrt{4c\sigma^4 - (x - \sigma^2 - c\sigma^2)^2}} dx, \quad j = 1, \dots, k \\ &= \frac{\sigma^2(1+c)}{2} - \frac{\sigma^2}{2\pi} \int_0^\pi (1+c - 2\sqrt{c}\cos\theta) d\theta \\ &= 0. \end{aligned}$$

Proof of (18) in main paper

Let $\underline{s}(z)$ be the Stieltjes transform of $(1 - c)1_{[0, \infty)} + cF_{c, \delta_1}$. One can show that

$$\underline{m}(z) = \frac{1}{\sigma^2} \underline{s}\left(\frac{z}{\sigma^2}\right).$$

Then, in Proposition 1, we have

$$v(f_j, f_l) = -\frac{1}{2\pi^2} \oint \oint \frac{f_j(\sigma^2 z_1) f_l(\sigma^2 z_2)}{(\underline{s}(z_1) - \underline{s}(z_2))^2} d\underline{s}(z_1) d\underline{s}(z_2), \quad j, l = 1, \dots, k. \quad (10)$$

For $g(x) = x$, we have

$$\begin{aligned} v(g) &= -\frac{1}{2\pi^2} \oint \oint \frac{g(\sigma^2 z_1) g(\sigma^2 z_2)}{(\underline{s}(z_1) - \underline{s}(z_2))^2} d\underline{s}(z_1) d\underline{s}(z_2) \\ &= -\frac{\sigma^4}{2\pi^2} \oint \oint \frac{z_1 z_2}{(\underline{s}(z_1) - \underline{s}(z_2))^2} d\underline{s}(z_1) d\underline{s}(z_2) \\ &= 2c\sigma^4, \end{aligned}$$

where $-\frac{1}{2\pi^2} \oint \oint \frac{z_1 z_2}{(\underline{s}(z_1) - \underline{s}(z_2))^2} d\underline{s}(z_1) d\underline{s}(z_2) = 2c$ is calculated in Bai et al. (2009) (it corresponds to $v(z_1, z_2)$, Section 5, proof of (3.4)).

Proof of (5)

By Proposition 1, for $\sigma^2 = 1$ and $g(x) = \log(x)$, by using the variable change $x = 1 + c - 2\sqrt{c} \cos \theta$, $0 \leq \theta \leq \pi$, we have

$$\begin{aligned} m(g) &= \frac{g(a(c)) + g(b(c))}{4} - \frac{1}{2\pi} \int_{a(c)}^{b(c)} \frac{x}{\sqrt{4c - (x - 1 - c)^2}} dx, \quad j = 1, \dots, k \\ &= \frac{\log(1 - c)}{2} - \frac{1}{2\pi} \int_0^\pi \log(1 + c - 2\sqrt{c} \cos \theta) d\theta \\ &= \frac{\log(1 - c)}{2} - \frac{1}{4\pi} \int_0^{2\pi} \log |1 - \sqrt{c} e^{i\theta}|^2 d\theta \\ &= \frac{\log(1 - c)}{2}, \end{aligned}$$

where $\int_0^{2\pi} \log |1 - \sqrt{c} e^{i\theta}|^2 d\theta = 0$ is calculated in Bai and Silverstein (2010).

Proof of (6)

By Proposition 1 and (10), for $\sigma^2 = 1$ and $g(x) = x$, we have

$$\begin{aligned} v(g) &= -\frac{1}{2\pi^2} \oint \oint \frac{g(z_1)g(z_2)}{(\underline{s}(z_1) - \underline{s}(z_2))^2} d\underline{s}(z_1) d\underline{s}(z_2) \\ &= -\frac{1}{2\pi^2} \oint \oint \frac{\log(z_1)\log(z_2)}{(\underline{s}(z_1) - \underline{s}(z_2))^2} d\underline{s}(z_1) d\underline{s}(z_2) \\ &= -2\log(1 - c_n), \end{aligned}$$

where the last integral is calculated in Bai and Silverstein (2010).

Proof of (8)

F_{c_n, δ_1} is the Marčenko-Pastur distribution of index c_n . By using the variable change $x = 1 + c_n - 2\sqrt{c_n} \cos \theta$, $0 \leq \theta \leq \pi$, we have

$$\begin{aligned} \int \log(x) dF_{c_n, \delta_1}(x) &= \int_{a(c_n)}^{b(c_n)} \frac{\log x}{2\pi x c_n} \sqrt{(b(c_n) - x)(x - a(c_n))} dx \\ &= \frac{1}{2\pi c_n} \int_0^\pi \frac{\log(1 + c_n - 2\sqrt{c_n} \cos \theta)}{1 + c_n - 2\sqrt{c_n} \cos \theta} 4c_n \sin^2 \theta d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{2 \sin^2 \theta}{1 + c_n - 2\sqrt{c_n} \cos \theta} \log |1 - \sqrt{c_n} e^{i\theta}|^2 d\theta \\ &= \frac{c_n - 1}{c_n} \log(1 - c_n) - 1, \end{aligned}$$

where the last integral is calculated in Bai and Silverstein (2010).

Proof of (7)

In the normal case with $\sigma^2 = 1$, Zheng (2012) gives the following equivalent expression of (3):

$$v(f_j, f_l) = -\lim_{r \rightarrow 1^+} \frac{\kappa}{4\pi^2} \oint \oint_{|\xi_1|=|\xi_2|=1} f_j(|1 + h\xi_1|^2) f_l(|1 + h\xi_2|^2) \frac{1}{(\xi_1 - r\xi_2)^2} d\xi_1 d\xi_2,$$

where $\kappa = 2$ in the real case and $h = \sqrt{c}$ in our case. We take $f_j(x) = \log(x)$ and $f_l(x) = x$, so we need to calculate

$$v(\log(x), x) = -\lim_{r \rightarrow 1^+} \frac{1}{2\pi^2} \oint \oint_{|\xi_1|=|\xi_2|=1} |1 + \sqrt{c}\xi_2|^2 \frac{\log(|1 + \sqrt{c}\xi_1|^2)}{(\xi_1 - r\xi_2)^2} d\xi_1 d\xi_2.$$

We follow the calculations done in Zheng (2012): when $|\xi| = 1$, $|1 + \sqrt{c}\xi|^2 = (1 + \sqrt{c}\xi)(1 + \sqrt{c}\xi^{-1})$, so $\log(|1 + \sqrt{c}\xi|^2) = \frac{1}{2} (\log(1 + \sqrt{c}\xi)^2 + \log(1 + \sqrt{c}\xi^{-1})^2)$. Consequently,

$$\begin{aligned}
\oint_{|\xi_1|=1} \frac{\log(|1 + \sqrt{c}\xi_1|^2)}{(\xi_1 - r\xi_2)^2} d\xi_1 &= \frac{1}{2} \oint_{|\xi_1|=1} \frac{\log(1 + \sqrt{c}\xi_1)^2}{(\xi_1 - r\xi_2)^2} d\xi_1 + \frac{1}{2} \oint_{|\xi_1|=1} \frac{\log(1 + \sqrt{c}\xi_1^{-1})^2}{(\xi_1 - r\xi_2)^2} d\xi_1 \\
&= \frac{1}{2} \oint_{|\xi_1|=1} \log(1 + \sqrt{c}\xi_1)^2 \left(\frac{1}{(\xi_1 - r\xi_2)^2} + \frac{1}{(1 - r\xi_1\xi_2)^2} \right) d\xi_1 \\
&= 0 + i\pi \left(\frac{1}{(r\xi_2)^2} \frac{2\sqrt{c}}{1 + \frac{\sqrt{c}}{r\xi_2}} \right) \\
&= 2i\pi \frac{\sqrt{c}}{r\xi_2(r\xi_2 + \sqrt{c})}.
\end{aligned}$$

Thus,

$$\begin{aligned}
v(\log(x), x) &= \frac{1}{i\pi} \oint_{|\xi_2|=1} |1 + \sqrt{c}\xi_2|^2 \frac{\sqrt{c}}{\xi_2(\xi_2 + \sqrt{c})} d\xi_2 \\
&= \frac{1}{i\pi} \oint_{|\xi|=1} (1 + c + c(\xi + \xi^{-1})) \frac{\sqrt{c}}{\xi(\xi + \sqrt{c})} d\xi \\
&= \frac{1}{i\pi} \oint_{|\xi|=1} \left(\frac{\sqrt{c}(1+c)}{\xi(\xi + \sqrt{c})} + \frac{c}{\xi + \sqrt{c}} + \frac{c}{\xi^2(\xi + \sqrt{c})} \right) d\xi \\
&= 2(1 + c - (1 + c) + c + 1 - 1) \\
&= 2c.
\end{aligned}$$

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